

Highlights

UAV-TAlign: Scene-Level RGB-Thermal Alignment for Cross-FOV UAV Imagery

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- A scene-level RGB-thermal alignment framework is introduced for cross-FOV UAV imagery.
- Robust scene consensus and QA-aware verification convert pairwise evidence into retained transforms.
- UAV-TAlign-1K provides a focused public evaluation subset and scalable label-free protocol.
- Baselines and ablations show that scene-level reliability remains critical after pairwise saturation.

UAV-TAlign: Scene-Level RGB-Thermal Alignment for Cross-FOV UAV Imagery

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Abstract

UAV RGB-thermal alignment is a prerequisite for reliable aerial inspection, monitoring, and nighttime multimodal perception. In real cross-FOV aerial imagery, the operating question is not only whether two frames can be matched, but whether the resulting alignment can be trusted at the scene level. Modern multimodal matchers can already produce abundant pairwise correspondences, but viewpoint variation, cross-resolution sensing, thermal rendering changes, and low-light imagery make the resulting frame-level homographies uneven in quality. This highlights a critical gap between pairwise match availability and scene-level alignment reliability. We present UAV-TAlign, a scene-level RGB-thermal alignment framework that turns strong pairwise multimodal matching into reliable scene-level decisions through reliability-guided evidence selection, robust scene consensus, and QA-aware verification. We further package UAV-TAlign-1K, a public evaluation subset with a scalable label-free protocol for real UAV RGB-thermal alignment. Experiments against classical and modern off-the-shelf baselines, protocol-consistency analysis, and cumulative module ablations show that, once pairwise matching is largely available, the strongest gains come from robust scene consensus and QA-aware scene retention.

Keywords: RGB-thermal registration, UAV imagery, infrared imagery, multimodal alignment, scene-level reliability

1. Introduction

RGB-thermal alignment is becoming a foundational capability for UAV inspection, monitoring, and nighttime multimodal perception. Visible imagery provides semantic and structural detail, while thermal imagery remains informative under illumination changes, smoke, haze, and low-light

conditions. This complementarity has driven progress across multispectral pedestrian perception, thermal object detection, visible-thermal tracking, and UAV multimodal sensing [1, 2, 3, 4, 5, 6, 7, 8]. For these systems to operate as fused sensors rather than separate streams, however, RGB and thermal observations must be aligned reliably at the scene level.

Real UAV RGB-thermal alignment is substantially more demanding than standard same-view image matching. Aerial platforms introduce viewpoint drift, cross-FOV capture, rolling scale changes, and cross-resolution sensors; thermal images further introduce low-texture regions, grayscale or pseudo-color rendering, and strong day/night appearance shifts. These factors make individual frame-level homographies uneven: some pairs are strongly supported, while others require additional scene context before they should influence the final transform. The practical question is therefore not only whether a matcher can produce correspondences for one pair, but whether a system can decide which scene-level alignment should be trusted.

Recent pairwise matching has advanced rapidly. Transformer-based, dense, and multimodal matchers such as LoFTR, LightGlue, DKM, RoMa, XoFTR, and MINIMA have substantially improved correspondence availability under difficult appearance changes [9, 10, 11, 12, 13, 14, 15, 16]. Our experiments confirm this progress in the UAV RGB-thermal setting: strong off-the-shelf multimodal matchers already achieve near-saturated pairwise usability on UAV-TAlign-1K. This result creates a concrete tension: a method can approach saturated pairwise usability while producing only selective scene-level retention under a strict acceptance rule. Pairwise usability and scene-level canonical acceptance therefore measure different operating questions and should not be read as the same success criterion. The key step is to aggregate evidence across a scene, suppress unstable homographies, and verify whether the final transform should be retained.

This observation motivates a scene-level framing of the problem. We study UAV RGB-thermal registration as a reliability-centered alignment task rather than a collection of independent pairwise matches. Under this view, modern matchers are powerful evidence generators, but the alignment algorithm must still answer three system-level questions: which frames should contribute evidence, how should multiple homographies be fused into a scene consensus, and when is the resulting transform reliable enough to use? This framing is especially important for UAV data, where the final output should favor trustworthy scene-level geometry over indiscriminate acceptance.

We present UAV-TAlign, a scene-level UAV RGB-thermal alignment frame-

work designed around this decision layer. UAV-TAlign operates on top of modern pairwise matchers and uses reliability-guided evidence selection to obtain representative frame support, robust scene consensus to consolidate frame-level estimates, and QA-aware verification to retain scene-level transforms that satisfy strict quality criteria. In this sense, UAV-TAlign elevates strong pairwise multimodal matching into a deployable scene-level alignment decision.

We also package UAV-TAlign-1K, a public evaluation subset for real UAV RGB-thermal alignment. The subset contains 15 scenes and 500 RGB-thermal pairs spanning day, night, low-light, grayscale thermal, pseudocolor thermal, and wide/zoom scene conditions. The subset is accompanied by a scalable label-free protocol that evaluates pairwise usability, scene-level retention, and reliability diagnostics. Together, the data and protocol provide a practical and reproducible way to study scene-level alignment reliability under real UAV RGB-thermal conditions.

Our contributions are threefold:

- We introduce UAV-TAlign, a scene-level UAV RGB-thermal alignment framework with reliability-guided evidence selection, robust scene consensus, and QA-aware verification.
- We package UAV-TAlign-1K, a public cross-FOV UAV RGB-thermal subset with a scalable label-free evaluation protocol.
- We show through off-the-shelf baselines, reliability diagnostics, and cumulative ablations that robust scene consensus and QA-aware verification dominate once pairwise matching is largely available.

2. Related Work

2.1. Classical Image Registration

Classical image registration pipelines typically rely on hand-crafted local features followed by robust geometric fitting. Representative methods built on SIFT, SURF, ORB, BRISK, KAZE, or AKAZE remain attractive because they are simple, interpretable, and do not require task-specific training [17, 18, 19, 20, 21, 22]. In cross-modal settings, the appearance gap between visible and thermal imagery can reduce descriptor repeatability, especially when thermal frames contain low-texture or homogeneous regions. These methods nevertheless provide a strong geometric reference point for

understanding which parts of the registration problem are addressed by local feature extraction and which parts require higher-level scene reasoning.

These methods remain important baselines in our study because they represent the classical registration view of the problem: stable local features followed by robust homography fitting. Robust model estimation remains central in these pipelines, from RANSAC and multiple-view geometry to more recent variants such as MAGSAC and MAGSAC++ [23, 24, 25, 26]. Our results show that real UAV RGB-thermal scenes benefit from moving beyond a single-pair formulation toward scene-level evidence aggregation and reliability control.

2.2. Deep Pairwise Matching for Multimodal Registration

Recent dense and semi-dense matching methods have significantly improved pairwise geometric correspondence estimation. Learned local features have evolved from trainable detectors and descriptors such as SuperPoint, LF-Net, D2-Net, R2D2, DISK, S2DNet, HardNet, L2-Net, SOSNet, GeoDesc, ContextDesc, ASLFeat, ALIKE, and Key.Net [27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40], while match filtering and end-to-end correspondence reasoning have progressed through SuperGlue, NCNet, Patch2Pix, COTR, LoFTR, ASpanFormer, LightGlue, DKM, RoMa, Efficient LoFTR, OmniGlue, MINIMA, and XoFTR [9, 41, 42, 43, 10, 11, 12, 13, 14, 44, 45, 16, 15]. In particular, these methods are better at recovering useful correspondences in low-texture regions, repetitive structures, or substantial appearance shifts. Broad reviews of this transition from classical pipelines to trainable matching, together with practical benchmarking studies, provide additional context [46, 47].

Our work builds directly on this progress by treating modern matchers as high-capacity evidence generators. The empirical picture in UAV-TAlign-1K is that strong off-the-shelf multimodal matchers already provide very high pairwise availability. This moves the central question from correspondence generation alone to the conversion of frame-level estimates into stable and reliable scene-level alignment under realistic UAV operating conditions.

2.3. Visible-Thermal Registration and Evaluation

Visible-thermal registration has been studied under a range of assumptions, from same-view or near-same-view alignment to broader multimodal correspondence learning. Radiation-variation insensitive matching methods such as RIFT and deep alignment or stitching formulations such as

CLKN and UDIS indicate the breadth of existing registration formulations [48, 49, 50]. Many existing settings emphasize relatively controlled view-point changes or pairwise registration outcomes. By contrast, practical UAV RGB-thermal data frequently combine cross-FOV capture, cross-resolution sensing, thermal rendering variation, and low-light conditions. These factors make scene-level reliability a more central concern than in many standard visible-thermal evaluation setups.

UAV-TAlign-1K extends this evaluation landscape by packaging real cross-FOV RGB-thermal scenes together with scalable label-free diagnostics. The protocol measures scene consistency and relative alignment quality, allowing the study to focus on reliability-controlled scene-level alignment under practical UAV acquisition conditions rather than isolated pairwise correspondence counts.

2.4. UAV Multimodal Datasets and the Remaining Gap

UAV multimodal datasets have supported progress in detection, tracking, segmentation, and broader cross-modal perception, spanning aligned pedestrian and automotive benchmarks as well as RGB-thermal detection and tracking benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV [1, 2, 3, 4, 5, 6, 7, 8]. These resources have primarily advanced downstream perception tasks, while scene-level RGB-thermal alignment places additional emphasis on reliability-aware geometric decision making. In particular, a practical registration pipeline must decide not only how to align frames, but also when the scene-level evidence is strong enough to trust the resulting transform.

This is the gap our paper targets. Complementary to new matcher architectures, we focus on a practical scene-level registration setting for real UAV RGB-thermal imagery. Our contribution lies in a reliability-oriented alignment framework, a public subset package, and a label-free protocol that make scene-level RGB-thermal reliability directly studyable.

3. UAV-TAlign-1K and Evaluation Protocol

3.1. Dataset Scope

We package UAV-TAlign-1K as a focused public evaluation subset for real UAV RGB-thermal registration. The subset contains 15 scenes, 500 RGB-thermal pairs, and 1000 images in total. Each sample is organized as an RGB image paired with a thermal image captured from the same scene

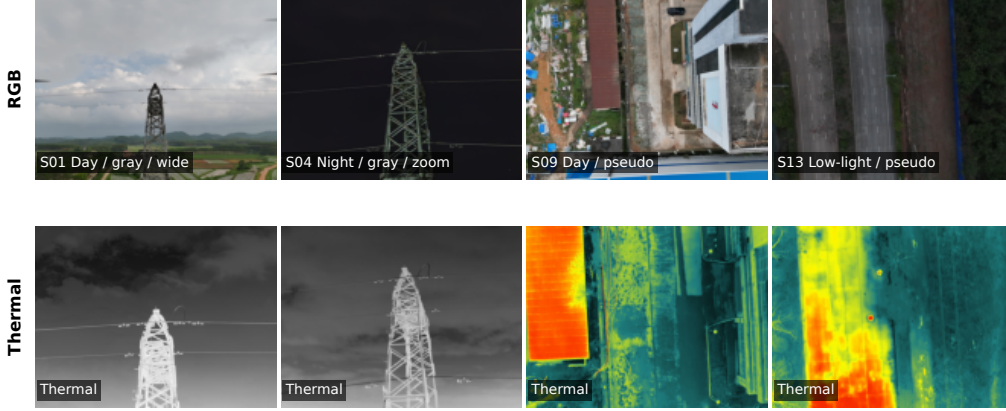


Figure 1: Representative UAV-TAlign-1K RGB-thermal pairs spanning day, night, low-light, grayscale-thermal, pseudocolor-thermal, wide-view, and zoom-view conditions.

context, while the dataset as a whole spans practical UAV conditions that go beyond standard near-same-view multimodal matching. In scope and modality, it is closer to recent RGB-thermal benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV than to classical same-sensor image matching setups [1, 2, 3, 4, 5, 6, 7, 8].

UAV-TAlign-1K is derived from the larger UAV-TAlign-12K collection, which contains 6039 RGB-thermal pairs and 12078 images across the same 15 UAV scenes. We use the 1K subset as the formal evaluation split in this paper so that all baselines, diagnostics, and ablations share a fixed, reproducible scene definition while preserving the key variation present in the larger collection.

At the scene level, the subset preserves the factors that make reliability assessment meaningful: day, night, and low-light capture; grayscale and pseudocolor thermal rendering; and both wide-view and zoom-view scene families. The packaged formal split includes 7 day scenes, 5 night scenes, and 3 low-light scenes; 8 grayscale-thermal scenes and 7 pseudocolor-thermal scenes; and paired wide/zoom controls for the substation scene family.

3.2. Dataset Organization

Each scene is stored under its own scene directory, and paired samples are formed by matching file stems between `rgb/` and `thermal/` subdirectories. The dataset loader constructs scene pairs from common RGB and thermal

filenames, while scene-level metadata are loaded from the subset manifest. For each pair, the loader stores scene identity, pair identity, light condition, thermal rendering, view type, and scene label.

This organization supports both pairwise baseline evaluation and scene-level pipeline evaluation under the same subset definition, while also enabling condition-based breakdowns from a unified scene manifest.

3.3. *Complementary Evaluation Units*

The protocol uses two complementary evaluation units. First, off-the-shelf baselines are evaluated at the pair level. Each of the 500 RGB-thermal pairs is processed independently, and the resulting statistics describe whether a method can produce a usable pairwise registration and homography under a fixed off-the-shelf setting.

Second, UAV-TAlign is evaluated at the scene level. Multiple frame-level estimates are aggregated into a scene-level transform, and the final decision is made by canonical scene acceptance rather than by pairwise usability alone. The pairwise and scene-level tables therefore provide two complementary views of practical registration behavior: correspondence availability and retained scene-level alignment reliability.

3.4. *Label-Free QA Proxies*

The protocol relies on label-free relative QA signals designed for scalable scene-level reliability analysis. The strongest stored proxies compare the optimized transform against a baseline initialization and summarize whether scene-level rectification improves cross-modal agreement. The proxy family includes edge-overlap improvement, gradient-NCC improvement, the ratio of frames with improved gradient agreement, severe baseline-relative outlier count and ratio, accepted-frame ratio, and homography-dispersion and robust-rejection diagnostics.

These proxies provide a practical scene-consistency signal for comparing methods and for deciding whether a scene-level transform is reliable enough to retain. They are intended for scalable reliability analysis and complement manual-landmark accuracy studies, in the spirit of practical matching-oriented evaluation and benchmarking discussed in prior image-matching literature [46, 47].

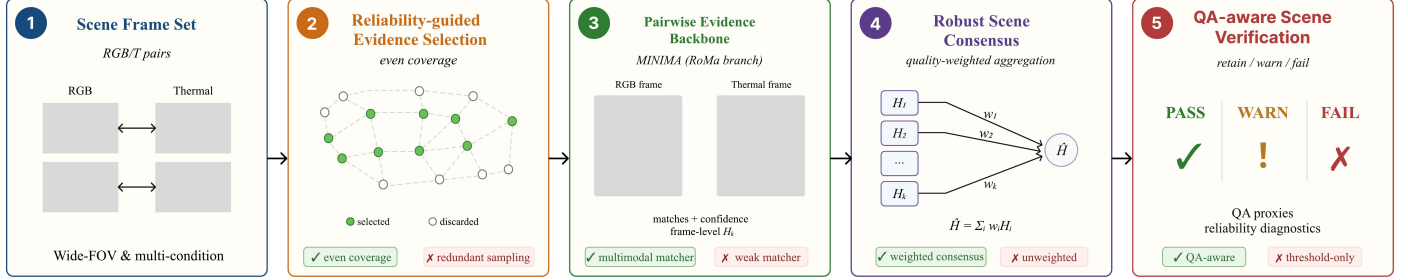


Figure 2: Overview of UAV-TAlign. The framework builds on strong pairwise matching, but makes the final decision at the scene level through reliability-guided evidence selection, robust scene consensus, and QA-aware verification.

4. Method

4.1. Problem Formulation

We study RGB-thermal registration as a scene-level alignment task. For each scene, the input is a set of paired RGB and thermal UAV frames $S = \{(I_i^R, I_i^T)\}_{i=1}^N$, where viewpoint variation, cross-resolution sensing, thermal appearance changes, and day/night conditions make individual frame estimates uneven in quality. The goal is to recover a thermal-to-RGB homography that remains reliable at the scene level, with the canonical output retained when the evidence satisfies the reliability criterion.

This formulation turns registration from a single best-pair problem into a multi-frame reliability optimization problem. In our setting, many scenes already contain usable pairwise correspondences, but the central challenge is to convert uneven frame-level estimates into a transform that is geometrically stable, internally consistent, and acceptable under a reliability-aware criterion.

4.2. Pairwise Evidence Backbone

UAV-TAlign uses a strong off-the-shelf multimodal matcher as its pairwise evidence generator. Our implementation uses the MINIMA family with the RoMa branch as the default backend, while the rest of the pipeline remains agnostic to the exact pairwise matcher [14, 16].

4.3. Reliability-Guided Evidence Selection

At the scene level, UAV-TAlign prioritizes the most informative frame evidence before expanding the support set. It uses a progressive evidence

schedule that expands with scene size and can cover all frames when needed. This reliability-guided policy allows the framework to stop when a compact subset already yields a stable consensus, while still escalating to broader evidence when the scene benefits from more support.

Within each stage, representative frames are selected with deterministic evenly distributed sampling by default. This default policy is used for reproducibility and stable scene coverage. A random-selection variant is reserved for sensitivity analysis and is reported separately.

4.4. Quality-Weighted Frame Geometry

For each selected frame pair, the matcher produces cross-modal correspondences that are filtered into a frame-level homography estimate. UAV-TAlign evaluates each frame using standard geometric statistics, including the number of matches, the number of inliers, inlier ratio, reprojection error, and spatial coverage. A frame enters the global candidate pool only when it passes a geometry-quality gate on these statistics. The overall geometric treatment remains rooted in standard homography estimation and robust fitting practice [24, 23, 25, 26].

Beyond a binary accept/reject decision, the pipeline assigns each accepted frame a quality score so that better-supported homographies receive more influence during scene aggregation. The score is a weighted combination of inlier ratio, spatial coverage, reprojection quality, matcher confidence, and inlier count. This keeps the scene-level estimate from being dominated by low-support frames that only marginally satisfy the frame-level gate.

4.5. Robust Scene Consensus

Once enough frame-level homographies have been accepted, UAV-TAlign forms a robust scene consensus. Homographies are mapped into parameter space, centered with a median reference, and filtered by a median-absolute-deviation rule to suppress unstable estimates before weighted averaging. The retained homographies are then combined using the frame quality scores defined above.

The pipeline records stability diagnostics around the aggregate transform, including homography dispersion relative to the aggregate and the ratio of robustly rejected frame-level estimates. These diagnostics connect consensus formation with the final QA-aware decision.

4.6. QA-Aware Scene Verification

The final decision integrates match availability with scene-level reliability evidence. UAV-TAlign evaluates the optimized scene-level transform against a label-free QA summary computed on representative frames. The QA summary uses cross-modal proxy signals such as edge-overlap improvement, gradient-NCC improvement, the ratio of improved frames, and the count of severe relative outliers. It also retains scene-level match and stability statistics such as accepted-frame ratio, mean inlier ratio, mean reprojection error, spatial coverage, dispersion relative to the aggregate transform, and the robust reject ratio.

The canonical decision uses a baseline-aware verification rule. A scene is retained when accepted-frame statistics remain strong, the aggregated homography is stable, severe baseline-relative outliers remain controlled, and the transform improves gradient-based alignment beyond configured floors. This final verification step gives UAV-TAlign its scene-level character: the framework prioritizes retained scene-level usability over unconditional scene acceptance.

5. Experimental Setup

5.1. Baselines

We evaluate both classical and learning-based baselines under an explicit off-the-shelf setting. All learning-based methods use public pretrained weights or public pretrained model interfaces and remain fixed on UAV-TAlign-1K. This choice keeps the comparison focused on practical deployability rather than dataset-specific adaptation. Unified inference wrappers standardize model loading, image I/O, homography fitting, and output schemas while preserving the public pretrained baseline parameters and settings.

The classical baselines are SIFT + RANSAC and AKAZE + RANSAC. The deep pairwise baselines are Kornia LoFTR (pretrained="outdoor"), RoMa outdoor full, XoFTR-640, and raw MINIMA. Raw MINIMA uses the MINIMA RoMa branch as a direct pairwise baseline; reliability-guided evidence selection, robust scene consensus, and QA-aware verification are activated in the UAV-TAlign full framework [17, 22, 10, 14, 15, 16].

5.2. Shared Geometric Fitting

For the pairwise methods, the common downstream step is homography fitting from the predicted correspondences. After each matcher returns point

pairs and optional confidence values, the shared evaluator estimates a homography with a robust backend and then records a unified outcome category. This unified post-processing is important because it makes the pairwise baseline table comparable at the level of usable registration outcomes rather than raw matcher outputs alone.

The homography fitting backend uses the same configuration across pairwise baselines:

- robust fitting method: USAC-MAGSAC;
- reprojection threshold: 3.0;
- confidence: 0.999;
- maximum iterations: 10000;
- spatial coverage grid: 4×4 .

5.3. Main-Table Baseline Choices

The main-table baseline choices are as follows. Raw MINIMA uses the MINIMA RoMa backend with the `minima_roma` checkpoint family and the large/full RoMa branch. RoMa uses the public outdoor full model. XoFTR uses the 640-resolution visible-thermal checkpoint family as the default main-table setting. Kornia LoFTR (pretrained="outdoor") uses the public pretrained outdoor model from Kornia.

5.4. Evaluation Details

The LoFTR baseline is evaluated under a resize-aware inference setting. The longest image dimension used for matching is limited to 1600, and automatic mixed precision is disabled. This deployment setting preserves the public pretrained model while supporting consistent full-subset evaluation within the fixed deployment budget.

All experiments were run in a Linux Conda environment with Python 3.10, PyTorch 2.0.1, torchvision 0.15.2, and CUDA 11.8. All evaluation outputs are written to isolated result roots, with raw dataset directories and source trees treated as read-only inputs during evaluation.

Table 1: Pairwise baseline comparison on 500 RGB-thermal pairs. Red/orange/yellow cells mark the best, second, and third values for each metric; arrows indicate the preferred direction.

Method	OK / N $\uparrow$	OK (%) $\uparrow$	H (%) $\uparrow$	Inlier $\uparrow$	Coverage $\uparrow$	Reproj. $\downarrow$	Runtime $\downarrow$
raw MINIMA	500/500	100.0	100.0	0.378	0.996	1.707	1.859
RoMa outdoor full	500/500	100.0	100.0	0.400	0.982	1.672	1.077
Kornia LoFTR outdoor	499/500	99.8	99.8	0.096	0.919	1.411	0.549
XoFTR-640	494/500	98.8	98.8	0.151	0.951	1.610	1.204
AKAZE + RANSAC	479/500	95.8	95.8	0.177	0.646	0.393	2.368
SIFT + RANSAC	457/500	91.4	91.4	0.240	0.599	0.260	4.605

Table 2: Formal scene-level retention result for UAV-TAlign. Arrows indicate the preferred direction.

Method	Eval Unit	Retained Scenes $\uparrow$	Retention Rate (%) $\uparrow$	Mean Accepted Ratio (%)
UAV-TAlign	15 scenes	7/15	46.7	64.5

6. Results

6.1. Pairwise Baseline Comparison

Off-the-shelf multimodal baselines already saturate pairwise usability on UAV-TAlign-1K. Table 1 shows a highly saturated pairwise regime: raw MINIMA and RoMa both reach 500/500 usable pairwise registrations, LoFTR reaches 499/500 under a resize-aware inference setting, and XoFTR-640 reaches 494/500. This result sharpens the role of UAV-TAlign. The framework targets the decision layer that remains after strong pairwise matching is available.

The secondary fitting statistics should be read as complementary diagnostics under a unified downstream estimator, rather than as a single absolute ordering of matcher quality across heterogeneous correspondence regimes.

6.2. Scene-Level Performance of UAV-TAlign

High pairwise usability does not automatically imply broad scene retention, because the canonical decision answers a different operating question. UAV-TAlign retains 7 of 15 scenes with a mean accepted-frame ratio of 64.5%. This operating point emphasizes high-confidence transforms and shows how robust consensus and verification turn abundant frame-level matches into selected alignment products.

Table 3: Scene-level breakdown by light condition. Red/orange/yellow mark the best, second, and third values within each metric.

Condition	Scenes	Retained $\uparrow$	Retention Rate (%) $\uparrow$	Mean Accepted Ratio (%) $\uparrow$
Day	7	4	57.1	77.1
Night	5	1	20.0	50.6
Low-light	3	2	66.7	58.5

6.3. Condition Breakdown

Retained alignments show meaningful condition structure, with night scenes remaining the most selective regime. Day and low-light groups retain multiple alignments, while the night group highlights the value of condition-aware analysis. These patterns support the view that the central challenge is decision quality rather than pairwise matcher availability alone.

6.4. Protocol and Proxy Evidence

QA-filtered decisions are more strongly associated with QA-relative outlier signals than with simple matcher availability. The baseline-relative QA outlier signal separates all 8 filtered scenes from the 7 retained scenes. Accepted-frame ratio is also informative as part of the joint decision: scene 02 has 98% accepted frames but is filtered because reject and QA outlier signals remain high, whereas scene 04 is retained with 80% accepted frames when these signals are absent. These observations show that UAV-TAlign captures transform usability as a joint QA decision rather than a simple count of matched frames.

6.5. Cumulative Ablation

The strongest module-level gain is produced by robust scene consensus, while deterministic selection is best understood as a reproducible operating policy. Adding robust scene consensus substantially improves edge and gradient alignment proxies while reducing severe outliers. Adding QA-aware verification then converts accepted-frame evidence into a reliability-controlled scene-level criterion, matching the formal 8-scene slice at 5/8 retained scenes.

The completed random-selection supplement further shows a 4/8 to 7/8 spread across seeds on the same fixed subset. Deterministic-even selection therefore serves as a reproducible default operating policy, while robust scene consensus and QA-aware verification remain the primary empirical drivers.

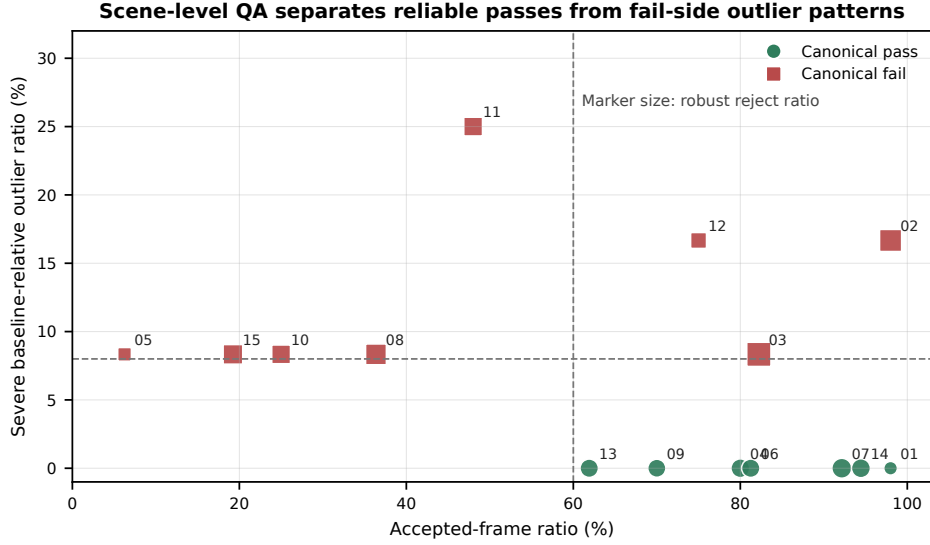


Figure 3: Scene-level proxy evidence. Retained scenes cluster near low severe-outlier ratios, while QA-filtered scenes are associated with stronger baseline-relative QA outlier signals. Marker size indicates robust rejection ratio.

Table 4: Cumulative module ablation on the fixed 8-scene subset. Red/orange/yellow mark the best, second, and third values for proxy metrics; arrows indicate preferred direction.

Variant	Evidence	Consensus	Verification	Retained/N $\uparrow$	Acc. $\uparrow$	Δ Edge $\uparrow$	Δ Grad $\uparrow$	Severe $\downarrow$
Evidence selection only	✓	–	–	8/8	77.4	0.124	0.126	10.4
+ Robust scene consensus	✓	✓	–	8/8	77.4	0.145	0.148	4.2
+ QA-aware verification	✓	✓	✓	5/8	77.2	0.147	0.149	4.2
UAV-TAlign full (8-scene slice)	✓	✓	✓	5/8	80.4	0.147	0.149	4.2

7. Discussion

7.1. Why Scene-Level Reliability Becomes Central

The results position UAV-TAlign as a scene-level reliability framework for UAV RGB-thermal alignment. Modern pairwise matchers already provide abundant frame-level correspondences on UAV-TAlign-1K, which makes scene-level decision quality the central operating question. UAV-TAlign addresses this layer directly by elevating strong pairwise correspondence models into reliability-controlled scene-level alignment products.

The protocol evidence supports this reliability framing. Retained and QA-filtered scenes show structured differences in accepted ratio, severe outlier behavior, and baseline-relative QA signals, indicating that the canonical

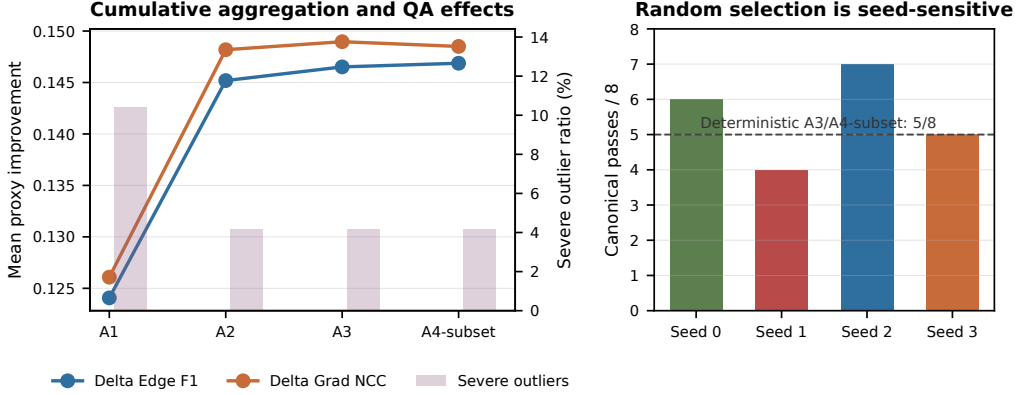


Figure 4: Cumulative ablation and random-selection supplement. Robust scene consensus improves proxy quality and reduces severe outliers, while the random-selection variant shows non-trivial seed sensitivity on the same fixed 8-scene subset.

criterion captures scene-level reliability beyond pairwise matcher availability. This gives the evaluation a clear diagnostic role: it identifies when abundant pairwise evidence becomes a trustworthy scene-level transform. The canonical scene criterion is designed for conservative retention of scene transforms, not for maximizing unconditional scene acceptance.

The completed multi-seed random-selection supplement further strengthens the design rationale for deterministic evidence selection: it provides a stable operating policy under the same fixed scene subset, while robust scene consensus and QA-aware verification deliver the strongest module-level gains.

7.2. Scope and Validity

UAV-TAlign targets real UAV scenes where a dominant planar or far-field transform is a useful operating model. Within this scope, public subset packaging, label-free diagnostics, and scene-level QA provide a compact experimental foundation for studying RGB-thermal alignment beyond isolated pairwise matching. This positioning is deliberately complementary to matcher-centric research: as correspondence models continue to improve, reliability control becomes increasingly important for deployable RGB-thermal systems.

The protocol is designed for scalable reliability analysis and is not presented as a substitute for manual geometric ground truth. Its QA proxies operate as scene-consistency diagnostics, while manual landmark accuracy

remains a complementary evaluation target. The homography model is most appropriate when a dominant planar or far-field alignment is meaningful, and it is less suited to strong parallax or highly non-rigid scene structure. UAV-TAlign-1K is therefore a focused public subset for this operating regime and provides a foundation that can be extended toward broader UAV RGB-thermal settings.

8. Conclusion

This work reframes UAV RGB-thermal registration from pairwise match availability to scene-level alignment reliability. UAV-TAlign turns strong off-the-shelf multimodal matching into retained scene-level decisions through reliability-guided evidence selection, robust scene consensus, and QA-aware verification. Across formal evaluation, protocol-consistency analysis, and cumulative module ablations, the evidence shows that the decisive step is transforming abundant frame-level matches into trustworthy scene-level alignment products.

Together with UAV-TAlign-1K and the accompanying label-free evaluation setup, UAV-TAlign provides a practical and reproducible foundation for studying when scene-level alignment is reliable enough to retain in real UAV RGB-thermal imagery.

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